

Sardar Vallabhbhai National Institute of Technology, Surat

Mid Semester Examination September 2025
Branch: Electronics and Communication Engineering
Subject: Digital Communications (EC 301)

Session: 2025-26

Semester: 5th



Time Duration: 1 Hour 30 Minutes

Maximum Marks: 30

- Note: 1. All the questions are compulsory.
2. Marks distribution for each question is clearly provided in the 'Marks' column.

Sl.	Questions	CO	Marks
1.	What is Inter Symbol interference (ISI)? Why it occurs? Discuss Nyquist's first criteria to achieve Zero ISI.	CO1	1+1+2
2.	Design the Unipolar NRZ, Unipolar RZ, Polar NRZ, Polar RZ, Bipolar and Manchester line coding schemes for the following sequence of bits: 1 0 1 1 0 0 1 Which of the above methods can be used for error detection and which methods can be used for time synchronization?	CO1	3+2
3.	Design the block diagram of a digital communication system. Explain the work of each block.	CO1	3
4.	Design the waveforms of ASK, FSK and PSK for the following sequence of bits: 1 0 1 1 0 0 1	CO1	2
5.	Design the block diagram of BPSK modulation and Demodulation. Explain the work of each block.	CO1	3
6.	Draw a figure of a duobinary signal for the following sequence: 1 0 1 1 0 1 Explain how the same sequence will be decoded at the receiver from the received duobinary signal.	CO1	2+1
7.	Probability of occurrence distribution of five symbols are given as follows: $P(A) = 0.40$, $P(B) = 0.20$, $P(C) = 0.15$, $P(D) = 0.15$, $P(E) = 0.10$. Construct the codes according to Shannon Fanno coding for these symbols. Calculate the Average code length, Entropy of the source, Efficiency and Redundancy. Further check whether the Kraft's inequality is being satisfied or not.	CO4	5
8.	For the above question, instead of Shannon Fano coding, use Huffman coding and calculate the Average code length, Entropy of the source, Efficiency and Redundancy. Further check whether the Kraft's inequality is being satisfied or not.	CO4	5